

# A Novel Model of the Spread of Emerald Ash Borers in Pennsylvania

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## 1 Introduction

The emerald ash borer (EAB) is a species of beetle native to parts of Asia that lays its eggs in the bark of ash trees. When the larvae hatch, they eat the inner bark of the tree, making it unable to transport water and nutrients between parts of the tree [5]. In its native environment, the emerald ash borer colonizes ash trees that are already weakened or damaged by another disease. In North American forests, though, the common species of ash do not have any resistance to emerald ash borers, and an infestation can occur in healthy trees, and is often fatal [7]. Consequently, in North America, emerald ash borers are considered highly invasive, as they are capable of infesting healthy trees, decimating the ash population of entire swathes of forest [11].

This is an issue because ash trees are an integral part of the environment in many parts of North America. Mature ash trees are a canopy species, providing shade for species in the understory. Loss of the canopy creates openings for invasive species that can out-compete native species that need shade. Additionally, ash trees provide food and shelter for many species of animals. Ash has also long been used by Indigenous peoples. Black ash in particular is a very valuable basketry material to many nations [5]. Ash trees are also used as a source of lumber, firewood, and as landscaping trees. The loss of ash trees in forests across America has significant ecological and economic consequences [8].

There are two main methods by which the emerald ash borers spread. First, the beetles carry themselves by their wings, which allow them to travel several kilometers per year. Second, humans carrying infested ash wood, such as firewood or nursery stock, inadvertently move the beetles far afield [7]. Different strategies are needed to mitigate each type of travel. In particular, using insecticides or introducing parasitic wasps that target the EAB may be used to alleviate problems locally. Quarantines, which ban the transport of firewood and lumber to and from infested regions or require such products to be heat-treated can help to reduce transregional spread, as can as well as education campaigns

about the dangers of emerald ash borers [7] [19].

In this paper, we explore the overall dynamics of both the short and long-distance spread of the beetles using the example of the State of Pennsylvania, and how each type of spread contributes to the expansion of the beetle’s population. In particular, we develop a novel model that includes both modes of spread, by combining a dynamical model for local forests with a stochastic simulation for transregional travel. By varying the probability of human transport of infested firewood in our model, we investigate the impact of lowering long-distance transport (for instance, with quarantines or education campaigns) on the spread rate of emerald ash borers. We find that even without human transport of emerald ash borers, the beetles eventually infect every region of Pennsylvania, but that human transportation significantly accelerates that spread.

## 2 Models and Methods

We model a region by representing it with a graph, where each node represents a small area, and each edge represents the adjacency of these subdivided areas. The dynamics within nodes are discussed in Section 2.1, while the dynamics between nodes are discussed in Section 2.2

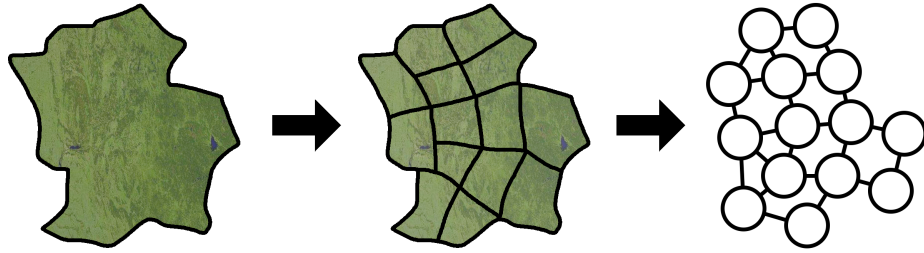


Figure 1: Turning a region into a graph

### 2.1 Local Spread

We start by constructing a system of differential equations to model the dynamics of emerald ash borers and ash trees within a single node. Here  $X$  represents the population of trees and  $Y$  represents the population of beetles.

$$\begin{aligned} X' &= A(X) - B(X, Y) \\ Y' &= C(X, Y) - O(Y) + I(Y) \end{aligned} \tag{1}$$

This system of equations is composed of five parts:  $A(X)$  describes the natural population dynamics of trees without beetle interference;  $B(X, Y)$  describes the detrimental effects of

the beetles on the trees;  $C(X, Y)$  describes the population dynamics of the beetles;  $O(Y)$  describes the number of beetles which move out of the node to its neighbors; and  $I(Y)$  describes the number of beetles which move into the node from its neighbors. How and why we arrived at this form of the system of equations shall be discussed further in the discussion section.

While the destruction of large numbers of ash trees may reduce pollination success, the regeneration of ash forests do not show a significant Allee effect, due to the fact that the beetles only target mature trees [2] [21]. Thus, we describe the natural population of trees with a logistic term:

$$A(X) = \rho X \left(1 - \frac{X}{K}\right) \quad (2)$$

The parameter  $\rho$  describes the growth rate of ash trees, and the parameter  $K$  describes the carrying capacity of the node. Due to our simplified model of ash tree growth, it is difficult to determine the appropriate value for  $\rho$ . Since the life cycle of ash trees is on a much longer time scale than the life cycle of emerald ash borers, this term is not very relevant for our model. For this project we let  $\rho = 0.1 \text{ yrs}^{-1}$  since it gives biologically reasonable dynamics. As for  $K$ , it shall be discussed in a later section.

Since the beetles depend entirely on the ash trees to survive during their larval stage, the population of ash trees is effectively a carrying capacity for the beetles. Thus, we also use a logistic function to describe the beetle population.

$$C(X, Y) = \beta Y \left(1 - \frac{Y}{QX}\right) \quad (3)$$

$\beta$  describes the reproduction rate of the beetles, and  $Q$  describes the number of beetles that a singular ash tree can sustain. A single female beetle lays between 60 and 90 eggs over the course of being an adult [13]. For the sake of simplicity we take this number to be 80 eggs. We assume that half of the beetles are female, and half of the larvae survive to adulthood. Thus, we want to find the value of  $\beta$  such that the population increases 20-fold after a year. To find an approximation, we suppose the population of trees is much greater than the number of beetles, i.e.  $\frac{\beta Y^2}{QX} \ll \beta Y$ . Then, we simply need to solve the boundary value problem  $Y'(t) = \beta Y(t)$  with  $Y(0) = 1$  and  $Y(1) = 20$ . This gives us  $\beta \approx 3 \text{ yrs}^{-1}$ .

As for  $Q$ , McCullough and Siegert [10] found that an infested tree produces roughly 100 adult emerald ash borers per meter squared of bark surface area. A mature ash tree has a height of up to 80 ft (24 m) and a trunk diameter of roughly two feet (0.6 m) [15]. Approximating a tree's bark area as a cylinder yields a surface area of roughly 90 meters squared, and thus a maximum of roughly  $Q = 9000$  ash borers per tree.

Finally, we hypothesize that the effect of beetles on trees decrease as the number of beetles increases. If 100 larvae were laid in an uninfested tree, they will have a much greater

impact on the tree's health than if they were laid in a tree that already has 500 larvae attacking it. Thus, we use a logarithmic term in the  $B(X, Y)$  function to model this fact:

$$B(X, Y) = \alpha X \ln(1 + Y) \quad (4)$$

The parameter  $\alpha$  describes the death rate of infested trees. Holden [4] and Knight, Brown, and Long [16] found that nearly all ash trees die within 6 years of becoming infected. We find that a value of  $\alpha = 0.05 \text{ yrs}^{-1}$  yields the described result (see Figure 2).

As for the terms  $O(Y)$  and  $I(Y)$ , which represent beetles moving out of and into the node, respectively, they shall be discussed in the next section.

Putting it all together, we have the following system of differential equations.

$$\begin{aligned} X' &= \rho X \left(1 - \frac{X}{K}\right) - \alpha X \ln(1 + Y) \\ Y' &= \beta Y \left(1 - \frac{Y}{QX}\right) - O(Y) + I(Y) \end{aligned} \quad (5)$$

This system is not able to capture the spread of beetles over long distances, since it assume all beetles and trees can instantly interact with each other. For this reason, this model must be combined with the long distance model discussed in the next section to accurately describe a large region.

Finally, due to the non-deterministic nature of the  $O(Y)$  and  $I(Y)$  terms, we choose to discretize the dynamical system in the simulation of our model. For simplicity, we use the forward Euler method, with a simulation step size of  $\Delta t$  years, yielding the following system of difference equations:

$$\begin{aligned} X_{t+1} &= X_t + \Delta t \left[ \rho X_t \left(1 - \frac{X_t}{K}\right) - \alpha X_t \ln(1 + Y_t) \right] \\ Y_{t+1} &= Y_t + \Delta t \left[ \beta Y_t \left(1 - \frac{Y_t}{QX_t}\right) - O(Y_t) + I(Y_t) \right] \end{aligned} \quad (6)$$

The behavior of the system is as follows: When there are no beetles, the trees maintain a population of  $K$ . When beetles are introduced, their population grows exponentially until it is capped by the number of trees. This causes the trees to quickly die out, which leads to the beetle population declining as well (see Figure 2).

## 2.2 Natural Transregional Spread

We apply the preceding model in each node in our graph. Next, we shall discuss the flow of beetles along the edges of our graph.

From Valenta et al. [23], emerald ash borers have been recorded to spread between 2.5 and 80 kilometers per year. On average, [3] estimates that they spread at an rate of 20 kilometers per year. In other words, they take an average of 0.05 years to travel a kilometer, however, they could take as short as 0.0125 years or as long as 0.4 years. Since there is a high variance in the amount of time it takes the beetles to travel a kilometer, we employ a stochastic model to simulate the inter-node spread.

For simplicity, we model each node as a perfect circle with no obstacles that may slow down the beetles' travel. We further assume that the beetles start at the center of the circle and travel in discrete steps, with each step being 1 kilometer long. Thus, for a node with a radius of  $r$  kilometers, the beetles will have to take  $r$  steps to spread throughout the node. Finally, we assume that the beetles have equal probability of taking a step at any moment in time. This suggests that if  $T$  is a random variable describing the number of simulation steps it takes for the beetles to travel 1 kilometer, then it must be that  $T$  is distributed according to the geometric distribution. Perhaps it would be more accurate to suggest that  $T$  is distributed according to the exponential distribution, however, since our computer simulation uses discrete time steps, we must use the exponential distribution's discrete counterpart: the geometric distribution. Thus, we want:

$$\mathbb{E}(T) \cdot \Delta t = \frac{1}{p} \cdot \Delta t = 0.05 \text{ yrs}$$

From this, we may find the probability of the beetles traveling 1 kilometer on each simulation step:

$$p = 20\Delta t$$

The aforementioned  $O(Y)$  term describes this stochastic process. For each node  $N$  with radius  $r_N$ , the amount of time it takes for the beetles to spread throughout a node is given by the negative binomial distribution. Let  $T_N$  be the random variable describing the number of simulation steps it takes for the beetles to spread throughout a node with radius  $r_N$ . We note the expected number of years before beetles have reached the edge of a node makes sense:

$$\mathbb{E}(T_N) \cdot \Delta t = \frac{r}{p} \cdot \Delta t = \frac{r}{20\Delta t} \cdot \Delta t = 0.05r \text{ yrs}$$

For each possible outcome  $\omega$ , we define  $O(Y_t)$  as follows:

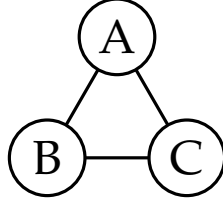
$$O(Y_t) = \begin{cases} 0 & \text{when } t < T_N(\omega) \\ \eta Y_t & \text{when } t \geq T_N(\omega) \end{cases} \quad (7)$$

That is to say,  $O(Y_t)$  "turns on" after the beetles have reached the edge of the node. Of course,  $\eta$  depends on the size and shape of each node, but for simplicity we shall simply let  $\eta = 0.1$ .

The number of beetles that leave each node is evenly sent to all its adjacencies. This is the  $I(Y)$  term of the local model. For a certain node  $N$ , we define:

$$I(Y_{Nt}) = \sum_{M \in \text{adjacencies of } N} \frac{O(Y_{Mt})}{\text{number of adjacencies of } M} \quad (8)$$

Thus, for a region modeled by  $n$  nodes, we have a  $2n$ -dimensional dynamical system. For example, if we have three nodes arranged in the following manner:



The full dynamical system would be described by:

$$\begin{aligned} X_{At+1} &= X_{At} + \Delta t \left[ \rho X_{At} \left( 1 - \frac{X_{At}}{K_A} \right) - \alpha X_{At} \ln(1 + Y_{At}) \right] \\ Y_{At+1} &= Y_{At} + \Delta t \left[ \beta Y_{At} \left( 1 - \frac{Y_{At}}{QX_{At}} \right) - O(Y_{At}) + \left( \frac{O(Y_{Bt})}{2} + \frac{O(Y_{Ct})}{2} \right) \right] \\ X_{Bt+1} &= X_{Bt} + \Delta t \left[ \rho X_{Bt} \left( 1 - \frac{X_{Bt}}{K_B} \right) - \alpha X_{Bt} \ln(1 + Y_{Bt}) \right] \\ Y_{Bt+1} &= Y_{Bt} + \Delta t \left[ \beta Y_{Bt} \left( 1 - \frac{Y_{Bt}}{QX_{Bt}} \right) - O(Y_{Bt}) + \left( \frac{O(Y_{At})}{2} + \frac{O(Y_{Ct})}{2} \right) \right] \\ X_{Ct+1} &= X_{Ct} + \Delta t \left[ \rho X_{Ct} \left( 1 - \frac{X_{Ct}}{K_C} \right) - \alpha X_{Ct} \ln(1 + Y_{Ct}) \right] \\ Y_{Ct+1} &= Y_{Ct} + \Delta t \left[ \beta Y_{Ct} \left( 1 - \frac{Y_{Ct}}{QX_{Ct}} \right) - O(Y_{Ct}) + \left( \frac{O(Y_{At})}{2} + \frac{O(Y_{Bt})}{2} \right) \right] \end{aligned} \quad (9)$$

### 2.3 Human-Driven Transregional Spread

Transportation of infested lumber and firewood by humans from infested regions to non-infested regions is the major form of long-distance spread of emerald ash borers [22], [11]. In fact, emerald ash borer was initially introduced to North America by infested wooden shipping crates [12]. Since we are interested in the result of changing human impacts on the spread of emerald ash borers, this is extremely relevant to our model.

For our model, we assume humans may bring infested lumber from any node to any other node, and that humans have equal probability of moving lumber at any moment in

time. Thus, the probability that a node becomes infested due to humans is proportional to number of nodes that are already infested. Unfortunately, there is scant information on the rate at which humans bring infested lumber products between regions, due to the beetle larvae being hidden under the bark. Thus, we arbitrarily select a probability of 1% per year per node for the purposes of our model. In other words, on the  $t$ th simulation step, an uninfested node has the following probability of becoming infested due to humans:

$$q(t) = 0.01N(t)\Delta t$$

where  $N(t)$  is the number of infested nodes on step  $t$ . In our simulation, for each node on each simulation step we “flip a coin” with probability  $q(t)$  of “success”. For simplicity’s sake, we presume the number of beetles brought by humans is very small. In this way, they will have a negligible effect on the source node, which might have millions of beetles. However, they will have a significant effect on an uninfected destination node, since they perturb the  $Y$  component of the difference equation (5) away from the unstable equilibrium at 0. We implement this in the simulation by adding 10 beetles to the node on a “success”. We do not subtract beetles from a source node since the number of beetles moved is negligible.

## 2.4 Modeling Pennsylvania

We decided to use a map of Pennsylvania for our model, as this is where we go to school.

We model each county in Pennsylvania as a node. Two nodes are connected with an edge if their counties are neighboring on the map. We use the square root of the area of each county as the radius parameter of its node.

From National Park Service [17], ash trees have a density of roughly 20 trees/ha, or 2000 trees/km<sup>2</sup>. For this paper, we approximate  $K$ , the carrying capacity of ash trees of each county, by multiplying its area by the density of ash trees.

We decided to begin the simulation by dropping a few beetles in Beaver county, as this is where they were first detected in Pennsylvania [18].

## 2.5 Software

We implemented our simulations with Python3 and NumPy. We acquired data on Pennsylvania county boundaries at [6]. We created all our figures using Matplotlib with geopandas. Our code may be found on GitHub.

### 3 Results

The dynamics between trees and beetles within each node as described by equation (6) can be seen in the following graphs. These graphs describe the dynamics of a single node without influence from neighboring nodes or human transport, to show how the populations would behave in an isolated setting. Note that the right-hand graph uses a logarithmic scale for the y-axis.

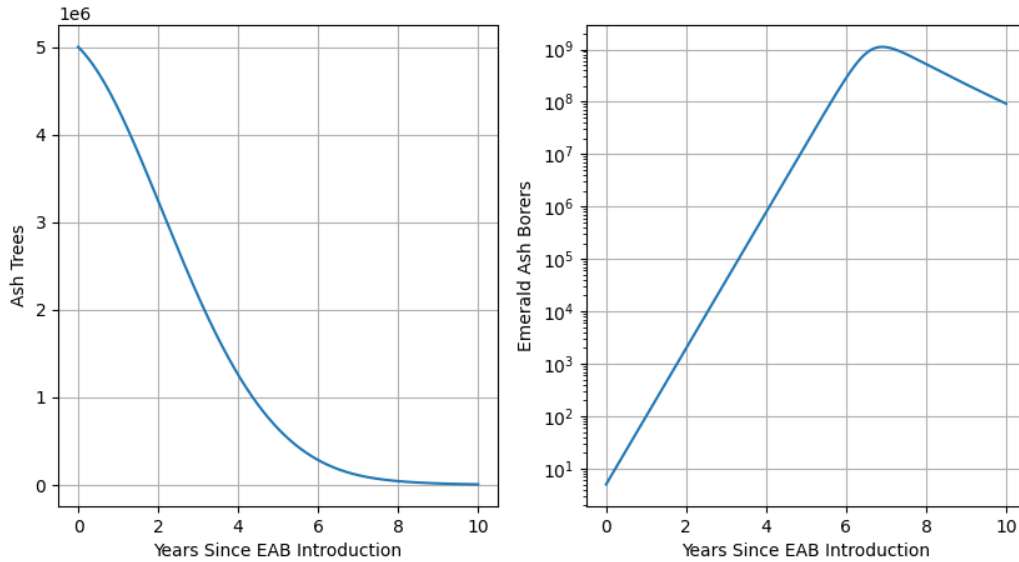


Figure 2: Graph of equation (6) with  $K = X_0 = 5000000$  trees and  $Y_0 = 5$  beetles for a duration of 10 years (with  $O(Y) = I(Y) = 0$ )

When beetles are introduced, the trees die off, while the beetle numbers grow exponentially until the tree population is so low that it can no longer support the number of beetles. 90% of ash trees die within 6 years after the introduction of beetles which matches the expected results for a single forest [4] [16].

#### 3.1 Qualitative Analysis

For our simulation of Pennsylvania, we introduce 10 beetles to Beaver county at time  $t = 0$  years, since the region around Beaver county was where emerald ash borers were first detected [18].

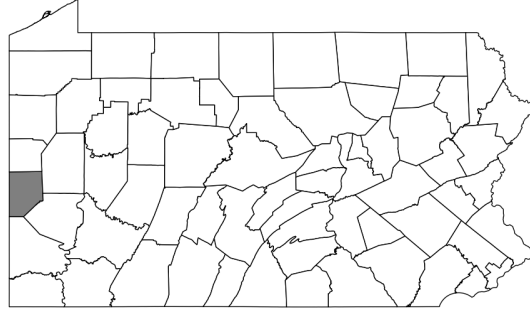


Figure 3: Location of Beaver county

Due to our implementing a stochastic term for the human-driven transregional spread of beetles, every time we run the simulation, the results differ slightly. Even though the stochastic terms cause each simulation to give different results, all of the counties are infected in around 10 years.

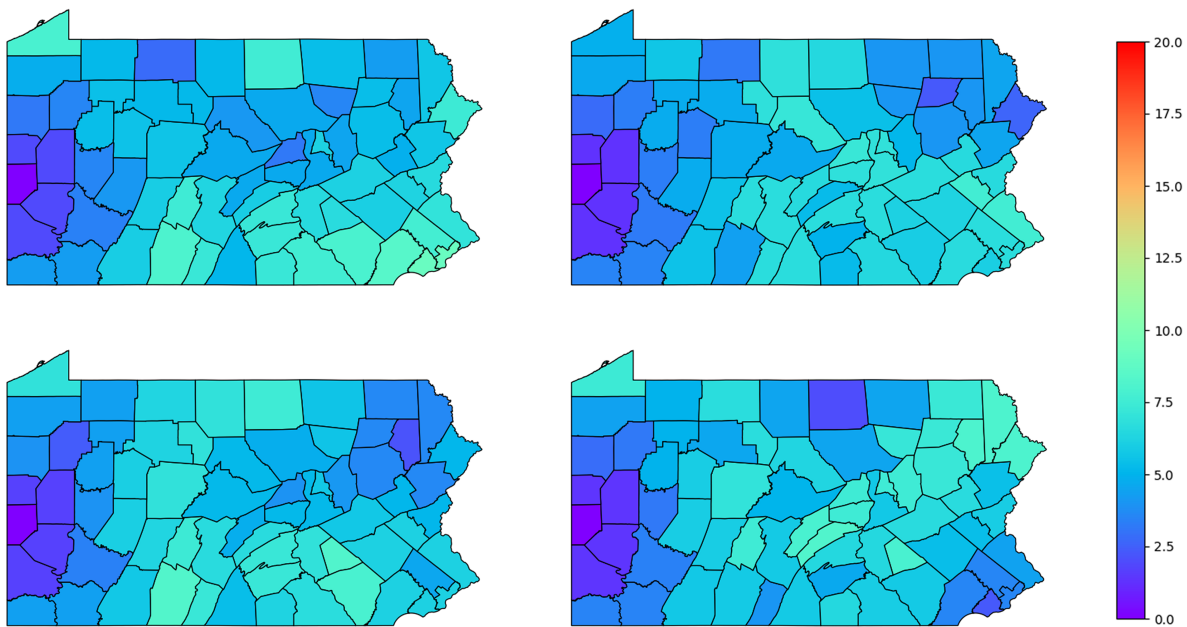


Figure 4: Several simulations with 1% probability of human-driven transport, color-coded by the number of years after initial infection it takes for EAB to appear (purple = 0 years, red = 20 years)

Now, in order to investigate the effect of human transport on the spread of firewood, we adjust the probability that a human will transport beetles between nodes. Examples with 0%, 0.01%, 0.5%, and 1% transport probability are shown in Figure 5.

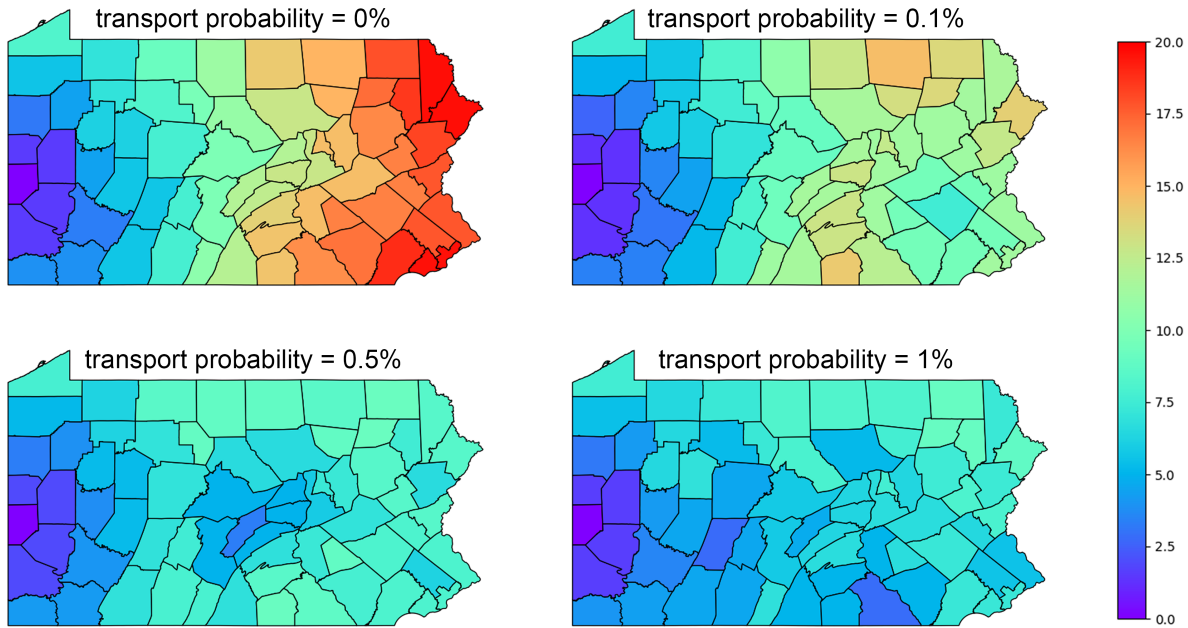


Figure 5: Several simulations with difference probabilities of human-driven transport, color-coded by the number of years after initial infection it takes for EAB to appear (purple = 0 years, red = 20 years)

We find that without human transport, the beetles spread relatively slowly across the region, resulting in complete infection after around 20 years. At a 0.01% probability of transport, total infection is reached at around 15 years, at a 0.5% probability in 12 years, and at a 1% probability in only 10 years.

### 3.2 Quantitative Analysis

To quantitatively analyze the spread rate of emerald ash borer given different levels of human activity, we measure the *mean time of first infection*, i.e. the average across every node of the number of years after initial infection it takes for EAB to appear. The mean time of first infection is a random variable, and it will only evaluate to a specific number given a specific simulation scenario.

We use the Monte Carlo method to analyze the mean time of first infection. For each transport probability, we run our simulation 200 times and calculate the mean time of first infection for each trial. We then estimate the true mean and true variance of the mean time of first infection using the sample mean and sample variance.

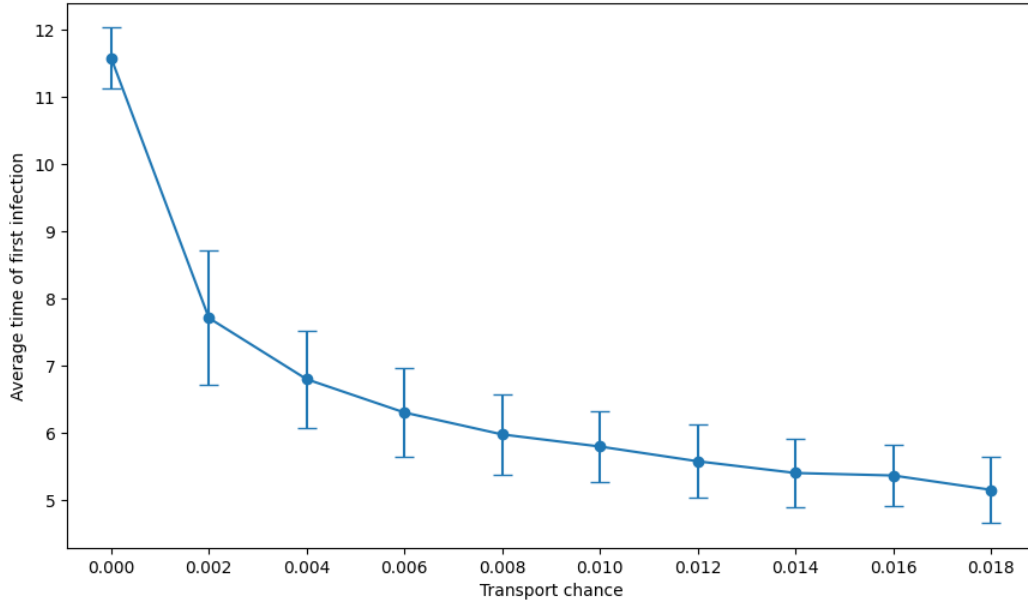


Figure 6: The mean and standard deviation of the mean time of first infection for select transport probabilities between 0% to 2%

## 4 Discussion

We find that the 1% human-driven transport case matches real-world data the most closely. Beetles were first detected in the first few counties in 2007, and they were found in the last remaining uninfected counties in 2017 [18]. Considering that the beetles can wipe out the vast majority of mature trees in roughly six years [16], this projects that near-total mature tree mortality occurs in 26 years without human spread, and as quickly as 16 years with humans transporting beetles.

### 4.1 Model Limitations and Future Work

For our model, we made a number of assumptions that limit the model's accuracy. Some of these were geographic. For instance, we modeled each Pennsylvania county as a node. We recognize that counties are perhaps a bit too large to accurately model with our local model. For example, in our current setup, beetles may need nearly a year to cross the largest counties. However, due to the limited scope of this project, we shall for now use this inexact solution.

We also assumed that the tree density and distribution within each county was the same, which is not necessarily the case. Since the total area of Pennsylvania is 115,883 km<sup>2</sup>, we approximated a total of 232 million ash trees. This is less than the 308 million ash trees that Pennsylvania actually has [1]. While this approximation is inaccurate, it is sufficiently close to the actual value as to allow for our model to produce reasonable results. Future researchers with more data could model each forest in every county with an individual node, and change parameters based on their sizes and tree densities, in order to create an accurate model.

Finally, we assumed the human-driven spread of EAB had an equal probability of delivering beetles from a node to any other node. In reality, not every county has an equal probability of people transporting beetles between them. To make the model more accurate, one could research where infested wood products are most likely to be transported to and from.

Generally, given the relevant data, our model can be applied to any region by future researchers, allowing them to model the potential spread of emerald ash borers in said location. There is also the potential, with some adjustment of parameters, to apply our model to other parasitic species that have similar population dynamics.

## 5 Conclusion

From our model we can conclude that measures taken to decrease human-driven spread, like quarantining infected sites and heat treating lumber to kill insects, could potentially delay the mean time of first infection by almost 6 years for a region the size of Pennsylvania (Figure 6). If we reached zero human-driven transport, the time to infection for the counties farthest from the initial infection site could increase by 10 years (Figure 5). This could allow time for other interventions, such as planting new trees and introducing parasitoids, to be implemented. However, this would require dramatically increasing awareness of the dangers posed by emerald ash borers in the general population to prevent any transregional spread. Thus, we ask that our readers keep their firewood local, and spread the word about the dangers of emerald ash borers!

## 6 Acknowledgment

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## Appendix I: The Creature

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